ReSpec — Critical Review and Build Specification

Waste & Energy track, WeMakeDevs Environmental Hacks 2026. Prepared Oct 9, 2026.

Evidence tags: [V] verified from a source in Section 19 (secondary copies noted) · [A] assumption to confirm · [E] my estimate · [S] synthetic demo data · [U] not verified by me.

---

1. Critical verdict

GO — conditional. ReSpec is the strongest of the concepts we have compared because it sits on a regulation that is already in force, targets the demand side (where the evidence says the bottleneck is), and gives AWS real work (Textract + Step Functions). I rate the concept about 7.5/10. The 91/100 score in your proposal is self-assigned and I do not endorse it. No concept can be promised a win; the previous event had 16,000+ participants.

Why it is credible [V]

- C&D Waste Rules 2025 took effect April 1, 2026. Projects of 20,000 m² built-up area or more, and road projects, must use processed C&D waste. The Second Schedule target is 5% of total construction material in 2026-27, rising to 25%. Occupiers must prepare a 'waste utilisation plan' approved by the local authority — a document ReSpec can draft.
- Delhi generated 6,053.8 TPD of C&D waste in FY 2023-24 against 5,150 TPD of plant capacity (DPCC annual report).
- The bottleneck is demand and procurement: MCD reported only 14% of recycled products were utilised in 2024-25, citing outdated material standards, weak demand and logistics. April–Nov 2025: about 450,000 t of processed material sold privately vs about 80,000 t procured by government agencies.
- An IIT-Madras customer-discovery study found quality concerns and usage restrictions make direct sale of recycled aggregate unattractive; value-added products such as pavers are the better channel.

What is wrong or unsupported in the proposal (fix before building)

1. **Facility list.** DPCC writes the plant as 'Ranikhera (Mundka)' — Rani Khera and Mundka are one site. 'Mundka/DMRC' as a fifth plant is unverified. Sources disagree on four vs five plants (and a planned Okhla plant). Build facilities.json from the DPCC table, mark each entry [U] until checked, and never show stock.
2. **Carbon claims.** I could not open the cited LCA papers. Remove CO2 from the MVP. Report tonnes only.
3. **DJB tender rules** were not opened by me. Do not encode anything from that PDF until a teammate reads the actual clauses.
4. **The target arithmetic is the real problem.** IS 383:2016 caps RCA at 25% (plain) and 20% (reinforced, up to M25) of the aggregate, and only lean concrete reaches 100%. The 5% rule is a share of total construction material. So most compliance must come from non-structural lines (pavers, kerbs, blocks, filling, sub-base). ReSpec must compute this gap honestly — it is the product's insight, not a footnote.
5. **ReSpec does not fix outdated standards.** Say so. It finds the lines where current standards already permit recycled material.
6. **'Verified' means evidence attached and consistent, not independently audited.** Demo evidence is [S].

Competitors and gap

- Delhi Malba (MCD app, Sept 2026): debris pickup to authorised plants with journey tracking — supply side [V].
- Revitalyze: platform for scouting/procuring recycled aggregates from pre-demolition data [V, per Cemex Ventures; location not checked].
- Doctor Sand (CDE Asia): B2B managed marketplace for recycled sand/aggregates; company claims recycled products sell about 30% cheaper [V as company claim].
- CPCB C&D portal: registration, EPR, reporting [U — cited in your proposal; I did not open it].
- I found no tool that reads a BOQ and checks standard-permitted substitution. That is absence of evidence, not proof. Spend 15 minutes on GitHub and Google tonight; if one exists, lead with the compliance-gap calculator and delivery ledger.

Rubric view

- Idea & Impact: strong problem and hook; impact is modelled until deliveries are recorded. State that plainly.
- Built on AWS: strong if Textract and Step Functions visibly run in the video.

- Design & Usability: medium risk — a domain-heavy UI must be understandable in seconds.
- Execution: **highest risk** — PDF table extraction and rule correctness.
- Demo: good if the 'gap vs target' reveal lands in the first minute.

How it loses: PDF extraction eats Friday; a civil-engineer judge finds one unsafe recommendation; the video explains regulation instead of showing the product; AWS looks decorative.

---

2. Prioritised improvements

MUST (MVP)

1. **Compliance-gap calculator:** modelled share of total material that safe, cited rules allow vs the Second/Third Schedule target.
2. **Rules with status:** every line gets ELIGIBLE / CONDITIONAL / MANUAL_REVIEW / NOT_ALLOWED with a source URL and clause. Anything without a cited rule defaults to MANUAL_REVIEW.
3. **Human approval** on every recommendation; structural and unknown-grade lines can never be auto-approved.
4. **Four-state ledger:** ELIGIBLE → PLANNED → ORDERED → VERIFIED. Only VERIFIED is reported as achieved.
5. **Input fallbacks:** PDF via Textract; CSV/XLSX and a paste-rows path in the browser so PDF trouble never blocks the demo.
6. **Draft waste-utilisation plan export** (CSV + PDF) marked DRAFT — not a CPCB filing.

SHOULD

- Facility ranking by straight-line distance, with a 'confirm stock by phone' banner.
- Evidence upload with consistency checks (delivered ⩽ planned + 5%, duplicate slip numbers blocked).
- Optional Bedrock category suggestion behind a feature flag.

CUT

- CO2/energy numbers, marketplace, payments, live stock, CPCB submission, optimiser beyond greedy selection, image classification, chatbot, mobile app, multi-tenant admin.

---

3. Final pitch

ReSpec reads a construction BOQ, shows which lines current standards allow recycled C&D material for, how far that gets a project toward the 2025 Rules' target, and tracks delivery evidence so planned diversion is separated from verified diversion.

- **Primary user:** quantity surveyor / procurement engineer on a Delhi-NCR project (public works, road repair, campus, large building).
- **Secondary user:** estate manager or approving engineer who signs off.
- **Why now:** the rules started April 1, 2026; the first targets bind this financial year.
- **Honest limits:** decision support, not structural certification or a statutory filing.

---

4. Workflow and screens

```
Upload BOQ → Extract → Review/correct → Classify → Evaluate rules → Gap vs target
→ Approve/reject → Plan export → Mark ordered → Record delivery + evidence → Verified
```

#	Screen	Must show	Acceptance criterion
S1	New project	name, type (building/road/campus), built-up area, site location, FY	Applicability message shown for ≥20,000 m² or road
S2	Upload	PDF/CSV/XLSX/paste; sample BOQ button; progress	Status updates within 5 s of stage change
S3	Review table	lines, confidence colours, inline edit	Edits persist; low-confidence cells highlighted
S4	Recommendations	status badge, product, max share, source clause, assumptions, approve/reject	No recommendation without a source or MANUAL_REVIEW
S5	Gap vs target	target %, modelled %, denominator and assumptions, lines driving the result	Assumptions visible on screen

#	Screen	Must show	Acceptance criterion
S6	Plan and ledger	procurement rows, facility options, ordered, evidence upload, 4-state totals	VERIFIED changes only after evidence exists

Empty state: 'Load sample BOQ'. Failure state: extraction error offers CSV/paste. Edge cases: missing unit, unknown grade, thickness missing, duplicate item numbers.

---

5. Rules engine and safety design

Categories (10): backfill/screened soil; granular sub-base; lean concrete (<M15); plain concrete; reinforced concrete; masonry/plaster sand; pavers; kerbstones; tiles; ineligible/manual review.

Seed rules that have a source [V]

- **R-LEAN:** IS 383:2016 Table 1 — lean concrete (<M15): RCA coarse and fine up to 100%; RA coarse up to 100%.
- **R-PLAIN:** plain concrete: RCA up to 25% (coarse and fine); RA nil.
- **R-RCC:** reinforced concrete: RCA up to 20%, only up to M25; RA nil. Prestressed: manufactured aggregates not permitted (clause 4.2.2).
- **Condition (Note 2):** only one type of manufactured coarse and one type of manufactured fine aggregate in a structure.
- **R-DELHI-2016:** Delhi advisory — at least 2% recycled products in building works, 10% in road works, 5% for non-structural application (2016; confirm currency).
- **R-CPWD:** CPWD direction of 10–20% recycled materials in non-sensitive CPWD works in NCT Delhi [V from a CSE presentation; date unverified].
- **R-SCHEDULES:** Second Schedule (buildings ≥20,000 m²) 5% in 2026-27; Third Schedule (roads) 5% for 2026-28, 10% for 2028-30, 15% from 2030-31 [V, secondary summaries; confirm in the Gazette].
- Products MCD lists: tiles, pavers, CC bricks/blocks, kerb stones, drain covers, filling material [V].

Rules NOT yet sourced [U] — ship as MANUAL_REVIEW until a mentor or teammate adds the clause: sub-base and backfill specification limits, paver/kerb product standards, masonry sand, tiles. Do not guess limits.

Rule record: id, category, condition (grade range, structural flag), allowed_product, max_share, basis (mass/volume), status, source_url, clause, last_checked, reviewed_by.

Unit conversion policy

- Normalise to m3, m2, m, nos, kg, t. Unknown unit → MANUAL_REVIEW.
- Mass is computed only with an explicit density or thickness factor in conversions.json, each with source and an engineer-override field. Seed values are [E] and labelled; missing factor → show volume only and exclude from the gap numerator.
- m2 lines (pavers, tiles) need thickness parsed from the description or entered by the user.
- Concrete lines need an aggregate fraction derived from an assumed mix [A]; show it and let the engineer override.

Extraction-error controls: confidence below threshold forces review; quantity outliers (>10× median in the same unit) flagged; unit/qty swapped patterns flagged; nothing is evaluated until the user confirms the table.

---

6. AWS architecture

```
Amplify (React) ─▶ API Gateway (HTTP API) ─▶ Lambda API ─▶ DynamoDB
                          │                        │
                          └─ presigned PUT ─▶ S3 (boq/, evidence/, textract/)
                                                   │
Lambda start ─▶ Step Functions
   1 StartDocumentAnalysis (Textract, TABLES)
   2 Wait 5s → GetDocumentAnalysis (loop, paginate)
   3 Normalize Lambda (rows → lines)
   4 Classify Lambda (keyword/fuzzy; Bedrock behind flag)
   5 Persist → status NEEDS_REVIEW
CloudWatch Logs + execution history; Evaluate/plan/deliver run in API Lambda
```

Service	Responsibility	Why necessary	Failure / fallback
S3	BOQs, Textract JSON cache, evidence	durable evidence chain	none needed
Textract (ap-south-1)	table extraction from PDF	core document-to-data step [V region]	CSV/XLSX/paste path

Service	Responsibility	Why necessary	Failure / fallback
Step Functions	orchestrate async extraction	async Textract needs polling; shows visible execution graph	catch → status ERROR with message
Lambda	API, normalise, classify, rules, plan	all logic	timeouts set; idempotent handlers
API Gateway (HTTP API)	public API	front door	CORS to app origin only
DynamoDB	jobs, lines, decisions, ledger events	state machine of the ledger	on-demand billing
Amplify	host the web app	public URL	S3+CloudFront if blocked
CloudWatch	logs with jobId correlation	execution evidence	—
Bedrock	optional category suggestion	disambiguate odd descriptions	flag off; keyword mapper is the default

Skip: OpenSearch, SQS/SNS, SageMaker, Cognito (use a shared demo passcode; say so), EventBridge.

Permissions: one Lambda role per function; S3 access scoped to its prefix; Textract Start/Get only on the pipeline role; presigned URLs expire in 5 minutes; no public buckets; evidence images may contain personal data — keep demo evidence synthetic.

Cost [V/E]: Textract tables cost about $0.015/page; the free tier gives 100 table-pages/month for 3 months on new accounts. Cache Textract output by file hash; test with small PDFs; set an AWS Budgets alarm at $10. Other services are negligible at demo scale [E].

Limits [V]: synchronous AnalyzeDocument handles single pages; multipage needs async. Mumbai allows 100 concurrent async jobs.

Show judges/mentors: a live Step Functions execution graph, the Textract JSON object in S3, DynamoDB items moving through states, CloudWatch lines with the job id, and the Amplify URL.

Recommended stack: TypeScript monorepo — /web (React+Vite), /api (Lambda), /core (rules, conversions, gap calculator; shared by API and browser), /infra (AWS SAM), /data (rules.json, facilities.json, conversions.json), /eval.

---

7. API contracts (freeze tonight)

```
POST /jobs                  {projectName, projectType: building|road|campus,
                             builtUpAreaM2?, fy, site:{lat,lng}}
                            → 201 {jobId, uploadUrl, key}
POST /jobs/{id}/start       {source: pdf|rows, rows?:
[{itemNo,description,unit,qty,rate?}]}
                            → 202 {status: QUEUED}
GET  /jobs/{id}             → {status:
UPLOADED|EXTRACTING|NEEDS_REVIEW|READY|ERROR,
                               stage, error?, lines: [Line]}
PATCH /jobs/{id}/lines      {edits: [{lineId, field, value}]} → 200 {lines}
POST /jobs/{id}/confirm     → 200 {status: READY}
POST /jobs/{id}/evaluate    → 200 {recommendations: [Rec], gap: Gap}
POST /jobs/{id}/decisions   {recId, decision: APPROVE|REJECT, approver,
note?} → 200 {rec}
POST /jobs/{id}/plan        → 200 {planId, rows: [PlanRow], exportUrls: {csv,
pdf}}
POST /jobs/{id}/orders      {recId, poRef} → 200 {rec}
POST /jobs/{id}/evidence-url {recId, filename} → 200 {uploadUrl, key}
POST /jobs/{id}/deliveries  {recId, facilityId, quantityT, slipNo, date,
evidenceKey}
                            → 201 {deliveryId, state: VERIFIED|FLAGGED,
flags}
GET  /jobs/{id}/impact      → {eligibleT, plannedT, orderedT, verifiedT,
assumptions}

Line  = {lineId,itemNo,description,unit,qty,rate?,confidence,category?,
         categoryConfidence?,gradeParsed?,flags[]}
Rec   = {recId,lineId,ruleId,status:
ELIGIBLE|CONDITIONAL|MANUAL_REVIEW|NOT_ALLOWED,
         product,maxShare,modelledMassT?,conditions[],source:{url,clause},
         decision?,ledgerState,facilityOptions[{facilityId,distanceKm}]}
Gap   =
{applicable,targetPct,modelledPct,denominatorT,numeratorT,assumptions[],
         topLines[]}
Errors: {error:{code,message,field?}} with 400/404/409/500.
```

8. Data model (DynamoDB single table, on-demand)

PK	SK	Attributes
JOB#id	META	project fields, status, stage, createdAt, sourceKey, textractHash
JOB#id	LINE#nnn	Line fields, edit history
JOB#id	REC#nnn	Rec fields, ledgerState
JOB#id	DEC#nnn	recId, decision, approver, timestamp
JOB#id	DELIV#nnn	recId, facilityId, quantityT, slipNo, date, evidenceKey, flags
JOB#id	PLAN#nnn	rows, export keys

Rules, facilities and conversions live in /data as versioned JSON packaged with the Lambda (deterministic, reviewable in git). Ledger transitions are append-only events; totals are derived.

9. AI vs deterministic

Component	Use	Why	Failure handling
Textract	table extraction	irregular PDF layout	confidence + review; CSV fallback
Keyword/fuzzy classifier	line → category (default)	fast, explainable	unknown → MANUAL_REVIEW
Bedrock (optional)	category suggestion as strict JSON from a fixed enum	odd descriptions	reject unknown enum; never sets eligibility
Rules engine	eligibility, limits, gap	safety-critical, citable	missing rule → MANUAL_REVIEW

Component	Use	Why	Failure handling
Greedy selector	pick facility by distance, fill targets	simple, reproducible	none

The LLM never states a limit and never approves anything.

---

10. Datasets and source register

Keep /docs/SOURCES.md with: name, URL, date accessed, licence, what we used, and a verified/unverified flag. Minimum entries: C&D Rules 2025 (Gazette); IS 383:2016 Table 1; DPCC FY2023-24 report; MCD/DTE utilisation reports; Delhi 2016 advisory; CPWD direction; a real public BOQ (one only, with its licence/terms); one malformed BOQ; one synthetic mini-BOQ [S] for tests; facility coordinates with their source. Textract and any library go in THIRD_PARTY_NOTICES.md.

---

11. Impact calculations and honest claims

- **Denominator:** total modelled mass of construction material lines, excluding metal and wood lines [A — the rules exclude usable metal, wood and glass; confirm the exact base with a mentor].
- **Numerator (modelled):** Σ line mass × approved recycled share.
- **Ledger totals:** eligible ≥ planned ≥ ordered ≥ verified, in tonnes.
- **Applicability:** projects ≥20,000 m² built-up or roads [V]; otherwise show 'not mandatory under the 2025 Rules; Delhi advisories may apply'.
- **Distance:** straight-line km, labelled estimate [E].
- **Claimable:** modelled tonnes of recycled material eligible/planned under cited rules; verified tonnes only from recorded evidence.
- **Not claimable:** CO2 or energy savings, field-validated diversion, availability or price of any product, structural suitability.

---

12. Evaluation plan

Hand-label 50 lines from the demo BOQ (include 10 negative controls: prestressed, M30+ RCC, retaining walls, water-retaining, unknown grade).

- Extraction: cell accuracy for item, unit, quantity.
- Classification: precision/recall per category.
- **Safety gate:** zero ELIGIBLE or CONDITIONAL results on negative controls. Any failure blocks submission.
- Performance: median time upload → NEEDS_REVIEW.
- Usability: number of manual corrections needed on the demo BOQ.
- Report only numbers actually measured; list the labelled set and script in /eval.

---

13. Four-person allocation

Engineer	Owns	Deliverable and acceptance
E1 Frontend	S1–S6, fixtures-first, PDF/CSV export UI, browser XLSX/CSV parse	Full UI works against /fixtures JSON with no backend; screens meet Section 4 criteria
E2 AWS/backend	SAM template, S3, API, Step Functions, Textract, DynamoDB, Amplify, CloudWatch, budget alarm	One command deploys; contracts in Section 7 return fixture-shaped data; Textract run completes on the demo PDF
E3 Domain/rules	taxonomy, rules.json, conversions.json, facilities.json, /core evaluate + gap + ledger, unit tests, mentor review	⩾25 unit tests pass; every ELIGIBLE rule has a source; negative controls pass
E4 Extraction/QA/demo	normaliser, classifier, labelled set, eval script, README, SOURCES.md, video	Eval report generated by script; README lets a stranger deploy; video under 3:00

Shared rule: nobody changes a contract without telling all four; change the fixture first, then code.

---

14. Build plan (internal hard stop Oct 11 18:00; recheck the official time)

Oct 9 (tonight)

- Fresh public repo; first commit now (history must start on/after Oct 8). Licence, README skeleton, SOURCES.md, THIRD_PARTY_NOTICES.md, AI-tool disclosure file.
- Choose the one demo BOQ (prefer a public CPWD/PWD/DJB-style file you may legally use; check format — XLS vs PDF).
- Freeze Section 7 contracts and fixtures. Run Textract on the demo PDF once and cache the JSON.
- **Checkpoint C1:** fixtures + empty deployed API + Textract JSON for the demo file.

Oct 10 — DTU (8 AM–8 PM)

- 8–10: mentors — AWS architecture review; civil/environmental mentor reviews rules and denominators.
- 10–12: upload→extract→review deployed. **C2 at 12:00.**
- 12–16: evaluate, gap, approve, plan, ledger. **C3 at 16:00.**
- 16–18: clean-browser rehearsal; run evaluation; fix blockers only.
- 18–20: second mentor critique; **feature freeze 20:00 (C4).**

Oct 11

- 8–10 smoke test signed-out; 10–12 final eval numbers into README; 12–14 record; 14–15 edit to ⩽2:55; 15–16 upload to YouTube and test signed-out; **target submit by 16:00**; buffer to the official deadline. Builder Center blog drafted by E4 in parallel.

Deployment: `sam build && sam deploy --guided` (region ap-south-1), connect the repo to Amplify for /web, set env vars from .env.example, run /eval, store outputs in the README.

---

15. Three-minute demo

- 0:00–0:15 Problem: rules in force, Delhi plants exist, recycled products go unused (14% utilisation).
- 0:15–0:30 User: a procurement engineer with a BOQ and a target.
- 0:30–1:00 Upload the real BOQ; Textract extraction; fix one low-confidence cell.

- 1:00–1:35 Recommendations with source clauses; one line shown as MANUAL_REVIEW (structural) to prove safety.
- 1:35–2:00 **Gap reveal:** 'This BOQ can reach X% against the 5% target' (use the real computed value) and which lines drive it.
- 2:00–2:20 Approve, export the draft plan, mark ordered.
- 2:20–2:35 Record delivery with demo evidence [S, labelled]; verified tonnes change only now.
- 2:35–2:52 AWS: Step Functions run, S3, DynamoDB, CloudWatch.
- 2:52–3:00 Close.

---

16. Judge simulation and red team

Judge	Likely rejection	Response in the build
Environmental	'Planned is not diverted.'	Four-state ledger; only VERIFIED claimed; no CO2
AWS engineer	'AWS is decoration.'	Textract + Step Functions run live in the video; execution graph shown
Product	'Why would engineers trust it?'	Source clause per line, human approval, conservative defaults, visible assumptions
Civil engineer	'Unsafe substitution.'	Structural/unknown → MANUAL_REVIEW; negative-control gate; DRAFT labels
Adoption	'Plant stock isn't live.'	Capability and distance only; confirm-stock banner

Other failure modes: BOQ format surprises (CSV/paste fallback), Textract merged cells (review table), mentor disagrees with a rule (downgrade to MANUAL_REVIEW), another team builds BOQ tooling (lead with gap calculator + ledger).

---

17. Deliverable checklists

A. AWS mentors / team

- [ ] Architecture diagram (Section 6) and region (ap-south-1)

- [ ] List of running resources with names
- [ ] A successful Step Functions execution and a failed one with its catch path
- [ ] CloudWatch log lines with jobId
- [ ] IAM roles least-privilege summary; S3 not public; presigned-URL expiry
- [ ] Cost note: Textract per page, free tier, budget alarm, caching
- [ ] Questions to ask: Is async Textract + polling the right pattern or should we use SNS/EventBridge? Table extraction tips for merged cells? Bedrock model/region availability from Mumbai and whether to skip it? Anything in our IAM or CORS that looks risky? What AWS evidence do judges find most convincing?

B. Hackathon judges

Mandatory [V, official rules and judging page]

- [ ] Public repository with code written on/after Oct 8 (history matches dates)
- [ ] Demo video on YouTube, public or unlisted, under 3 minutes, showing AWS in use; test link signed out
- [ ] Short write-up: problem, what we built, where AWS fits
- [ ] Licences and credits for libraries, templates, APIs; AI coding tools disclosed
- [ ] Student verification completed by every member (rewards/interview wait on it)

Recommended (not stated as mandatory)

- [ ] Deployed app URL and demo login/passcode
- [ ] README with setup, architecture, limitations
- [ ] SOURCES.md and evaluation results with the labelled set
- [ ] Honest impact section separating modelled, planned and verified
- [ ] AWS Builder Center blog (prize category)

C. Engineering team

- [ ] Source code, monorepo structure, lint/test scripts
- [ ] Setup and deploy instructions; .env.example; SAM template
- [ ] API documentation (Section 7) and fixtures
- [ ] Tests: /core unit tests (≥25), normaliser tests, negative controls
- [ ] Dataset/source register; THIRD_PARTY_NOTICES.md; licences checked (LICENCE of our code, BOQ source terms)

- [ ] AI-tool disclosure file
- [ ] Known limitations: decision support only; modelled quantities; rules coverage; facility stock unverified; demo evidence synthetic

---

18. Action plan for the next 60 minutes

- **E1:** scaffold /web, build S3 and S4 from fixtures first.
- **E2:** create the AWS account resources via SAM skeleton; S3 bucket, HTTP API, Textract test on the demo PDF; budget alarm.
- **E3:** draft rules.json from the Section 5 sourced rules only; write facilities.json from the DPCC table; list rules needing a mentor.
- **E4:** create the public repo and first commit now; pick and label the first 20 BOQ lines; write the contracts into /fixtures; search GitHub for existing BOQ-to-recycled tools.

---

19. Sources

- Hackathon pages: wemakedevs.org/aws/env, /rules, /schedule (hours still being finalised when checked on Oct 9)
- C&D Rules 2025 summaries: corporateprofessionals.com, downtoearth.org.in, gazettetracker.com (Gazette text, 2nd Schedule 2026-27: 5%); official gazette link in your proposal: egazette.gov.in/WriteReadData/2025/262313.pdf
- DPCC FY2023-24 annual report (dpcc.delhi.gov.in, annual-report-cd-for-fy-2023-2024.pdf) — read via search snippet only
- Down To Earth (MCD 14% utilisation) and Indian Infrastructure, May 2026 (450,000 t vs 80,000 t)
- IS 383:2016 Table 1 via civilapps.in and slideshare copies; MoEF report on Delhi 2016 advisory; CSE presentation on CPWD direction
- IIT-Madras study (irepose.iitm.ac.in); Cemex Ventures (Revitalyze); NBM&CW (CDE Asia/Doctor Sand); The Tech Outlook (Delhi Malba)
- AWS: docs.aws.amazon.com/general/latest/gr/textract.html (regions, quotas); AWS News Blog (Textract free tier and pricing); extend.ai summary (current table price)